

Orso Forghieri

Researcher in Reinforcement Learning and Applied Mathematics

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Research Profile

Researcher working on Reinforcement Learning, Dynamic Programming, Markov Decision Processes, state abstraction, hierarchical methods, and scalable decision-making. PhD thesis: *Hierarchical Reinforcement Learning for Large Scale Problems*, prepared at École polytechnique / Institut Polytechnique de Paris, within CMAP, supervised by Erwan Le Pennec, Hind Castel-Taleb, and Emmanuel Hyon.

Experience

- **Oct. 2025-Mar. 2026** - Quantitative Researcher, Qube Research & Technologies, Paris. Worked on sequential and explainable decision-making methods for systematic equity trading, including large-scale market-data analysis and signal evaluation.
- **2022-2026** - PhD in Applied Mathematics / Reinforcement Learning, CMAP, École polytechnique, Institut Polytechnique de Paris. Developed scalable approximation, abstraction, and disaggregation methods for high-dimensional Markov Decision Processes.
- **2021-2025** - Research collaboration in Reinforcement Learning and Combinatorial Optimization, Orange Gardens / CMAP. Worked on latency-constrained service placement for edge-computing infrastructures in an industrial 6G research context.
- **Mar. 2022** - Research Intern in Hierarchical Reinforcement Learning, T?l?com SudParis, in collaboration with Energy4Climate. Explored server energy and performance optimization under uncertainty.
- **Mar. 2020** - Machine Learning Intern, SNCF. Modelled and forecasted delay propagation in the rail network with boosted decision trees and Bayesian networks.
- **Sep. 2019** - Research Project in Market Forecasting, Lusi Trading Solutions. Studied LSTM, CNN, and NLP methods for financial time series and news headlines.

Education

- **2021-2022** - MSc MVA, Mathematics, Vision, Learning, École normale sup?rieure Paris-Saclay.
- **2018-2021** - Dipl?me d'ing?nieur, École polytechnique.
- **2015-2018** - Classe pr?paratoire aux grandes ?coles MPSI-MP, Lyc?e Louis-le-Grand.

Selected Publications

- *Hierarchical Reinforcement Learning for Large Scale Problems*, PhD thesis, Institut Polytechnique de Paris, 2025.
- *Faster Latency Constrained Service Placement in Edge Computing with Deep Reinforcement Learning*, IFIP Networking 2025.
- *Approximate Dynamic Programming and State Abstraction Discovery*, manuscript under peer review.
- *State Abstraction Discovery from Progressive Disaggregation Methods*, EWRL 2024.
- *Progressive State Space Disaggregation for Infinite Horizon Dynamic Programming*, ICAPS 2024, pp. 221-229.

Teaching

- Teaching Assistant in Statistics, Engineering degree, École polytechnique, Sep. 2024 and Sep. 2023.
- Teaching Assistant in Probabilities, MSc, École polytechnique, Sep. 2024 and Sep. 2023.
- Teaching Assistant in Rare Events Numerical Simulation, Engineering degree, École polytechnique, Mar. 2024.
- Teaching Assistant in Operational Research, BSc, Universit? Paris Nanterre, Feb. 2023.

Academic Service

- Doctoral representative on the Laboratory Life Committee, CMAP, École polytechnique, 2024-2025.
- Reviewer for ROADEF 2025; alpha tester for the Marmote MDP solver.
- Mathematics interviewer for the MSc X-HEC Data Science for Business, Feb. 2024.
- Outreach speaker at IHES with Bachelor students from École polytechnique, Nov. 2023; member of CNRS GDR ROD working groups through MyGDR.

Skills

Python, NumPy, SciPy, JAX, PyTorch, TensorFlow, Pandas, Gym/Gymnasium, Git, R, Java, C++; reinforcement learning, approximate dynamic programming, state abstraction, Markov Decision Processes, stochastic optimization, network/resource optimization, statistical learning.